VARUNA

AI-Powered Underwater Marine Debris and Anomaly Detection

Mission Summary

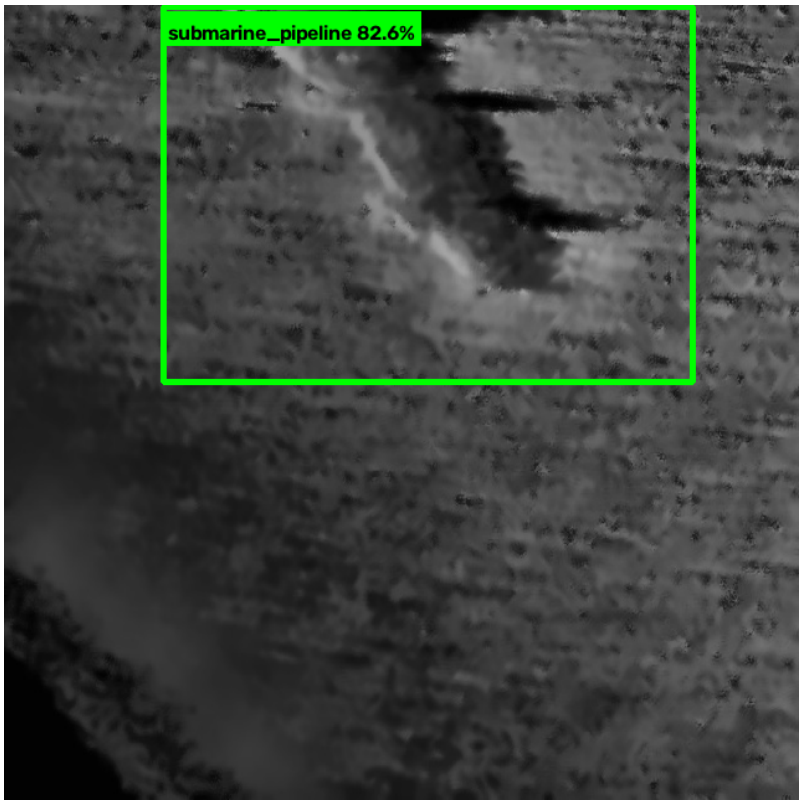
Frames processed	7
Accepted detections	6
submarine_pipeline	1
ghost_net	1
shipwreck	2
mine_cylinder	2

Geotagged Detections

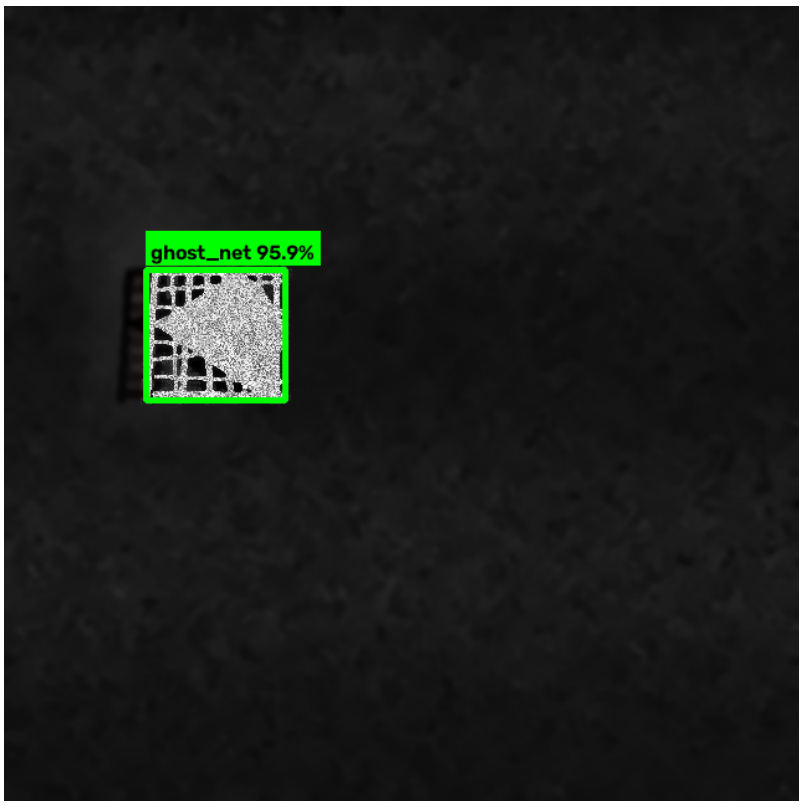
Class	Conf.	Latitude	Longitude	Depth	Event
submarine_pipeline	82.6%	15.396025	73.721333	35.0 m	3
ghost_net	95.9%	15.396425	73.722292	35.0 m	4
shipwreck	88.3%	15.396838	73.721840	35.0 m	5
shipwreck	60.0%	15.397940	73.723438	35.0 m	5
mine_cylinder	84.5%	15.396750	73.722896	35.0 m	6
mine_cylinder	80.3%	15.398289	73.723875	35.0 m	6

Annotated Sonar Frames

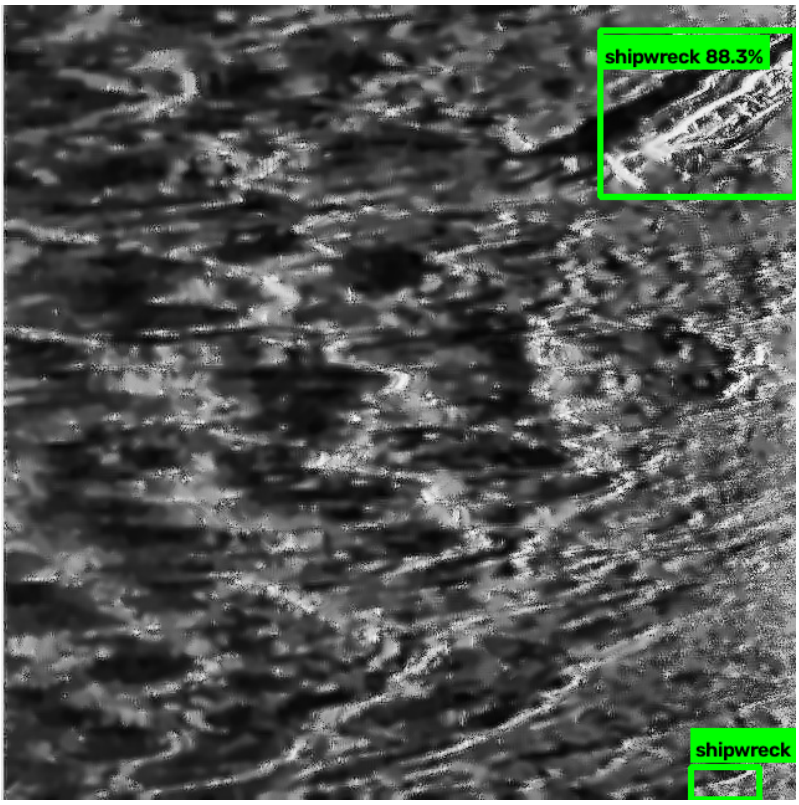
Event 3



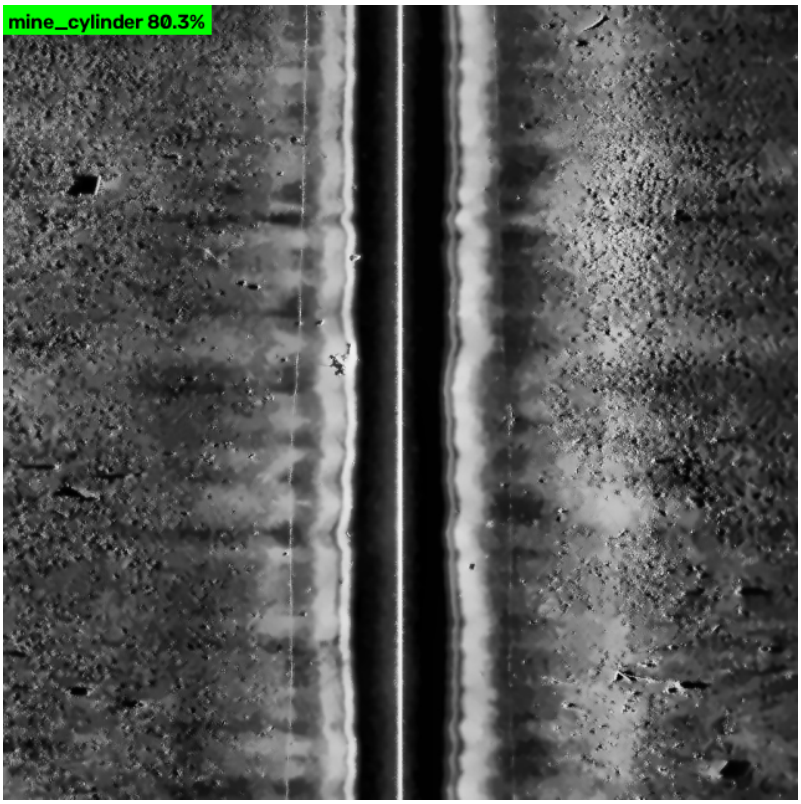
Event 4



Event 5



Event 6



Technical Notes

This demonstration uses a synthetic but structurally valid XTF mission. Navigation coordinates are demonstration data and are not survey-grade. Geotagging combines XTF ping navigation with sonar pixel geometry.